

SOBHAN FOOLADI MAHANI

AI RESEARCH ENGINEER | TELECOMMUNICATIONS & DEEP LEARNING

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SUMMARY

AI and Deep Learning Engineer with a strong foundation in Telecommunications Engineering and a recent MSc graduate with highest academic honors (10/10 Thesis). Passionate about bridging the gap between cutting-edge AI research and scalable, production-ready software applications. Brings comprehensive expertise in end-to-end development, from designing energy-efficient ML models, GenAI systems, and LLM-RAG pipelines to building robust Python and React-based full-stack platforms. Proven track record in reproducible benchmarking, agent-based orchestration, and optimizing data-intensive deep learning algorithms across heterogeneous edge and server hardware. Currently serving as a CTO, leveraging a multidisciplinary technical background to drive architecture strategy and deliver intelligent, highly optimized digital products.

TECHNICAL SKILLS

- **Programming & Core Languages:** Python, JavaScript, C++, Bash, HTML5/CSS3, SQL, Git, Linux (Debian/Ubuntu environments)
- **Deep Learning & AI Research:** PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib; CNNs, LSTMs/GRUs, transfer learning, contrastive learning, self-supervised learning, model evaluation
- **GenAI & NLP Systems:** Hugging Face, Llama models, multilingual NLP, transformer embeddings, LLM-RAG pipelines, vector stores, agentic workflows
- **Full-Stack & Backend Engineering:** React, JavaScript (frontend); Python-based backend frameworks (Django, Flask), REST APIs, asynchronous processing, application architecture, ingestion pipelines
- **Efficient AI & Edge Systems:** CodeCarbon, GPU/CPU/RAM profiling, hardware-aware benchmarking, resource-constrained optimization (edge vs. server)
- **Testing, Infrastructure & CI/CD:** Pytest, unit/integration testing, API testing; Docker, cloud endpoints, workflow orchestration, Sphinx/OpenAPI documentation

PROFESSIONAL EXPERIENCE

CTO & Full-Stack Developer

SpainU | Lanzadera (Marina de Empresas), Valencia

May 2026 – Present

- Architecting and developing a brand-new, end-to-end platform for the company as the technical leader.
- Building scalable Python backends and interactive, responsive React/JavaScript frontends.
- Designing application architecture, managing deployment workflows, and leading tech-stack decision-making within the Lanzadera startup accelerator ecosystem.

Master Thesis Researcher – Sustainable AI (SAI), Part time

CTTC – Centre Tecnològic de Telecomunicacions de Catalunya, Barcelona

Supervisors: Dr. Roberto Pereira, Dr. Paolo Dini

Oct 2025 – May 2026

- Conducted advanced research on energy-efficient deep learning, earning a perfect score of 10/10 on the Master's thesis presentation.
- Conducting research on energy-efficient deep learning, profiling training/inference costs and CO₂ emissions across heterogeneous hardware (Jetson, laptop GPUs, server nodes).
- Benchmarking CNN, LSTM/GRU, and contrastive learning models for image and time-series tasks under diverse training settings.
- Designing measurement campaigns using CodeCarbon.
- Deploying models on edge devices to study runtime constraints, memory limits, and batch-size effects.
- Optimization: Boosted MobileNet/EfficientNet stability on constrained hardware achieved 60%+ accuracy on CIFAR-100 with resource-aware training recipes.
- Designed reproducible scientific workflows for model benchmarking and resource profiling
- Developed an internal Python tool to automate ML training pipelines, experiment tracking, and result parsing for data-intensive benchmarking workflows across heterogeneous hardware. GitHub: [SustainVision](#)

AI & NLP Engineer, Freelance

Apr 2025 – Feb 2026

Moodest, Techno Campus, Mataro, Barcelona

- Developed multilingual LLM and NLP pipelines in Catalan, Spanish, and English for workplace well-being and communication analytics.
- Integrated Llama-based conversational components for multilingual human-machine interaction, including context handling, routing logic, and API-based inference services.
- Contributed to continuous model improvement through feedback-driven evaluation and pipeline refinement.
- Achieved 96% message-filtering accuracy and 91% engagement-score prediction using transformer embeddings and XGBoost models.
- Built privacy-compliant preprocessing pipelines and production-ready REST APIs for internal AI services.
- Implemented unit and integration tests with Pytest to support reliable deployment across cloud and edge environments.

AI & Backend Engineer (R&D – MAXSENS Project), Fulltime

Aug 2023 – Aug 2025

Imaz Tech, Barcelona

- Developed Flask APIs for real-time EMG and motion data processing, enabling scalable and modular AI pipelines.
- Built CNN and LSTM models achieving 94% gesture-classification accuracy on high-dimensional physiological and time-series signals.
- Improved signal-processing efficiency by 30% through optimized filtering and feature-extraction workflows.
- Automated analytics report generation in CSV, PDF, and JPG formats, reducing manual handling by 60%.
- Implemented posture and motion detection features that reduced exercise-execution errors by 40%.
- Collaborated with clinical and product teams to align ML outputs with real-world use cases.

Software Engineer / Technical Product Owner
SaminCoin, Tehran

Oct 2021 – Dec 2022

- Designed and maintained REST APIs in Python/Django for fraud detection and transaction classification.
- Created system architecture and UML diagrams to improve communication across teams.
- Contributed to roadmap planning and sprint execution in a cross-functional product environment.
- Worked with UI/UX teams to improve onboarding workflows for early users.

EDUCATION

MSc in Telecommunications Engineering (MATT) – Specialization in Deep Learning for Multimedia Processing **2023 – 2025**
Universitat Politècnica de Catalunya (UPC), Barcelona

Bachelor’s in Information Technology Engineering **2015 – 2020**
Azad University, Karaj

CERTIFICATIONS & INTERESTS

Languages: English (Fluent), Spanish (A2)
Interests: GenAI systems, efficient deep learning, edge AI, signal analysis, intelligent networked systems, chess